Representing and utilizing DDI in relational databases

A new DDI best practices working paper

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Agenda

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- Introduction
- Pros and cons of DDI in relational database systems
- Modeling DDI in relational databases
- Advanced cases
- Ensuring application compatibility
- An outlook to the future
- Q&A

Contributors

- The idea for this paper was formed at a workshop on mapping of DDI to relational databases in Frankfurt / Main in April 2011
- Contributors are:
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Introduction

- Modern research needs a good documentation for
  - reuse of data
  - data merging
  - international comparison of datasets
- DDI seems to be the most promising solution for standardized metadata documentation
- But DDI needs to be used practically (not only developed)

Introduction

- Therefore DDI must be easy to implement and proof for future developments in the areas of data storage and data analysis
- Relational databases are a widely used and flexible solution for data storage
- Bringing DDI together with the capability of relational database systems will promote both data storage for the purpose of scientific research and the DDI standard itself

- This presentation and the underlying paper outlines the advantages and disadvantages of representing DDI in relational databases as an alternative to an XML structure.

DDI in RDBs – pros and cons

- Pros of relational databases in regards to DDI
  - Structure is very good for rectangular files (e.g. SPSS or Stata)
  - Easier combination between metadata and microdata by using the same storage structure (e.g. by referential integrity)
  - Very common structure with high degree of optimization (e.g. indexes, file groups, stored procedures)
  - Capability to store multiple studies in one database system (more opportunity for harmonization between studies)
  - Internal independence of DDI version (can be adapted in the import and export processes on each individual version)

DDI in RDBs – pros and cons

- Pros of XML structures in regards to DDI
  - XML is native to DDI therefore no compatibility issues (e.g. unknown nodes do not have necessarily to be processed)
  - Hierarchical structure is difficult to model in relational databases
  - Full set of DDI leads to a very complex relational database with heavy response times due to complex joins (nevertheless most DDI-XML implementations only use a subset)
  - DDI-XML can easier be verified against the DDI schema
- An interesting approach is to use a hybrid relational database with XML acceleration or processing (e.g. enterprise databases like SQL Server or Oracle)

Modelling DDI in RDBs

- The paper does not include a model relational database using DDI or direct implementation examples, because there are too many surrounding factors to give a complete model, e.g.
  - Database engine (e.g. MySQL, Oracle, SQL Server)
  - Agency requirements (e.g. DDI elements needed)
  - Programming environment (e.g. PHP, Java, C#/.NET)
  - Previous database knowledge or structures within the agency
  - Old data which has to be migrated

- Therefore the paper is designed as a best practice guidebook derived out of the experiences in respective agencies

Modelling DDI in RDBs

- The paper includes the following design best practices:
  - DDI Elements
  - XML Hierarchie
  - References
  - Recursive structures
  - Substitution groups
  - Controlled vocabularies
  - Database Ids

Advanced Cases

- Versioning (including late bound references) can be established the following way in a relational database
  - Array of triggers on fitting tables
  - Managed code / external programming
  - Data warehouse technology (slowly changing dimensions)
- Modelling schemes which include another scheme
  - Model relational database very similar to DDI-XML structure
  - „Resolve“ all included schemes and only store the „complete“ version
- Two ways for multi language support
  - Exporting translations into XLIFF files (XML translation standard)
  - Direct injection from tables into DDI-XML files while exporting

Advanced Cases

- Handling unknown or external elements in DDI can be constructed in several ways, e.g.
  - RDB has a full set of DDI (therefore the problem does not occur)
  - Discarding unknown elements while importing the XML-DDI structure
  - RDB buffers unknown elements as strings or native XML (ideal solution in this case would be a database which can handle XML natively)

Ensuring application compatibility

- Improving DDI-XML import and export mechanism by use of DDI Profiles
- Topic is important for all DDI related exchange processes (e.g. also between DDI-XML databases)
- DDI Profile is a collection of XPaths that describe the objects within DDI that are either used or not used for particular purposes
- Use of a DDI Profile is not mandatory, but when one is being used, it should be referenced in all of the DDI instances that conform to it
- Paper includes an XML example of this structure
- Structure is very useful for communication of applications between or within agencies

An outlook to the future

- DDI does not need to rely upon a particular technical representation, but is valuable as an abstract model as can be seen from previous experiences
  - DDI 2 (until 2.5) was modeled as DTD
  - DDI 3 (all versions) are modeled as XSD
  - Many agencies support DDI as an import and export model, but internally use something different (e.g. relational databases or other repositories)
- Idea: the manifestation can be in different representations like UML or RDF
- Advantage: a technical representation can be generated out of the abstract model.
- Maybe a possible preparation for “DDI 4”?

The working paper

- The paper has been released on Friday, December 2nd, 2011 on the DDI website as part of the working paper series
- Please download it here: http://www.ddialliance.org/resources/publications/working/othertopics/RepresentingAndUtilizingDDIInRelationalDatabases.pdf
- DOI: http://dx.doi.org/10.3886/DDIOtherTopics02
- We would be happy for reviews, comments or other scientific discussions

Any Questions?

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